

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2024 Worked Solutions

January 2024 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2024

January 2024 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Polynomials

1. $f(x) = ax^3 + 3x^2 - 8x + 2$ where a is a constant

Given that when $f(x)$ is divided by $(x - 2)$ the remainder is 3, find the value of a .

(3)

Worked Solution - Question 1

1. Use the remainder theorem

Dividing by $(x - 2)$ means the remainder is $f(2)$.

2. Substitute x equals two

$$f(2) = a(2)^3 + 3(2)^2 - 8(2) + 2 = 8a + 12 - 16 + 2 = 8a - 2.$$

3. Use the given remainder

The remainder is 3, so $8a - 2 = 3$.

4. Solve for a

$$8a = 5, \text{ hence } a = \frac{5}{8}.$$

Final answer

$$a = \frac{5}{8}.$$

Question 2

Binomial Expansion

2. Find the coefficient of the term in x^7 of the binomial expansion of

$$\left(\frac{3}{8} + 4x\right)^{12}$$

giving your answer in simplest form.

(3)

Worked Solution - Question 2

1. Choose the x power

In $\left(\frac{3}{8} + 4x\right)^{12}$, the term in x^7 comes from choosing $(4x)^7$ and $\left(\frac{3}{8}\right)^5$.

2. Write the term

The coefficient is $\binom{12}{7} \left(\frac{3}{8}\right)^5 4^7$.

3. Simplify the powers

$4^7 = 2^{14}$ and $8^5 = 2^{15}$, so $\left(\frac{3}{8}\right)^5 4^7 = \frac{3^5}{2}$.

4. Evaluate

$\binom{12}{7} = 792$, so the coefficient is $792 \cdot \frac{243}{2} = 96228$.

Final answer

96228.

Question 3

Circles

3. The circle C

- has centre $A(3, 5)$
- passes through the point $B(8, -7)$

(a) Find an equation for C .

(3)

The points M and N lie on C such that MN is a chord of C .

Given that MN

- lies above the x -axis
- is parallel to the x -axis
- has length $4\sqrt{22}$

(b) find an equation for the line passing through points M and N .

(3)

Worked Solution - Question 3

1. Find the radius squared

The centre is $A(3, 5)$ and the circle passes through $B(8, -7)$, so
 $r^2 = (8 - 3)^2 + (-7 - 5)^2 = 25 + 144 = 169$.

2. Write the circle equation

Hence C has equation $(x - 3)^2 + (y - 5)^2 = 169$.

3. Use the horizontal chord

Since MN is parallel to the x -axis, its equation has the form $y = c$.

4. Use half the chord length

The chord length is $4\sqrt{22}$, so the half-length is $2\sqrt{22}$.

5. Relate chord length to the circle

At height $y = c$, the horizontal half-length satisfies
 $169 - (c - 5)^2 = (2\sqrt{22})^2 = 88$.

6. Solve for c

$(c - 5)^2 = 81$, so $c = 14$ or $c = -4$.

7. Use the position above the x -axis

The chord lies above the x -axis, so $c = 14$. Therefore the line through M and N is $y = 14$.

Final answer

(a) $(x - 3)^2 + (y - 5)^2 = 169$.

(b) $y = 14$.

Question 4

Integration

4. (a) Sketch the curve with equation

$$y = a^{-x} + 4$$

where a is a constant and $a > 1$

On your sketch show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote to the curve.

(3)

x	-4	-1.5	1	3.5	6	8.5
y	13	6.280	4.577	4.146	4.037	4.009

The table above shows corresponding values of x and y for $y = 3^{-\frac{1}{2}x} + 4$

The values of y are given to four significant figures, as appropriate.

Using the trapezium rule with all the values of y in the table,

- (b) find an approximate value for

$$\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx$$

giving your answer to two significant figures.

(3)

- (c) Using the answer to part (b), find an approximate value for

(i) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} \right) dx$

(ii) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx + \int_{-8.5}^4 \left(3^{\frac{1}{2}x} + 4 \right) dx$

(3)

Worked Solution - Question 4

1. Find the y-intercept

For $y = a^{-x} + 4$, set $x = 0$: $y = a^0 + 4 = 5$, so the curve meets the y -axis at $(0, 5)$.

2. Find the asymptote

Since $a > 1$, $a^{-x} \rightarrow 0$ as $x \rightarrow \infty$, so the horizontal asymptote is $y = 4$.

3. Describe the sketch

The curve is decreasing, passes through $(0, 5)$, and approaches the line $y = 4$ from above.

4. Use the trapezium width

The x values increase by 2.5, so $h = 2.5$.

5. Apply the trapezium rule

$$\int_{-4}^{8.5} (3^{-x/2} + 4) dx \approx \frac{2.5}{2} [13 + 4.009 + 2(6.280 + 4.577 + 4.146 + 4.037)].$$

6. Round part b

This gives 68.86125, so to two significant figures the estimate is 69.

7. Remove the constant part

$$\int_{-4}^{8.5} 4 dx = 4(12.5) = 50.$$

8. Estimate part c i

Using the answer to part (b), $\int_{-4}^{8.5} 3^{-x/2} dx \approx 69 - 50 = 19$.

9. Use symmetry by substitution

For $\int_{-8.5}^4 (3^{x/2} + 4)dx$, let $u = -x$. This changes it to the same integral as part (b).

10. Estimate part c ii

Therefore the two integrals in part (c)(ii) are approximately equal, so the total is $69 + 69 = 138$.

Final answer

(a) y -intercept $(0, 5)$ and asymptote $y = 4$.

(b) 69.

(c)(i) 19.

(c)(ii) 138.

Question 5

Sequences & Series

5. (i) Find the value of

$$\sum_{r=1}^{\infty} 6 \times (0.25)^r$$

(3)

- (ii) A sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 3$$
$$u_{n+1} = \frac{u_n - 3}{u_n - 2} \quad n \in \mathbb{N}$$

- (a) Show that this sequence is periodic.

(2)

- (b) State the order of this sequence.

(1)

- (c) Hence find

$$\sum_{n=1}^{70} u_n$$

(2)

Worked Solution - Question 5

Topic group

1. Identify the geometric series

$\sum_{r=1}^{\infty} 6(0.25)^r$ has first term $6(0.25) = 1.5$ and common ratio 0.25 .

2. Sum to infinity

$$S_{\infty} = \frac{1.5}{1 - 0.25} = \frac{1.5}{0.75} = 2.$$

3. Find the first few terms

$$u_1 = 3. \text{ Then } u_2 = \frac{3 - 3}{3 - 2} = 0.$$

4. Continue the recurrence

$$u_3 = \frac{0 - 3}{0 - 2} = \frac{3}{2}, \text{ and } u_4 = \frac{\frac{3}{2} - 3}{\frac{3}{2} - 2} = 3.$$

5. Show periodicity

Since $u_4 = 3 = u_1$, the terms repeat as $3, 0, \frac{3}{2}, 3, 0, \frac{3}{2}, \dots$

6. State the order

The repeating cycle has length **3**, so the order is **3**.

7. Sum one cycle

$$\text{One full cycle sums to } 3 + 0 + \frac{3}{2} = \frac{9}{2}.$$

8. Use 70 terms

Since $70 = 23 \cdot 3 + 1$, the first **70** terms contain **23** complete cycles and one extra term $u_1 = 3$.

9. Find the required sum

$$\sum_{n=1}^{70} u_n = 23 \left(\frac{9}{2} \right) + 3 = \frac{207}{2} + \frac{6}{2} = \frac{213}{2}.$$

Final answer

(i) 2.

(ii)(a) Periodic.

(ii)(b) Order 3.

(ii)(c) $\frac{213}{2}$.

Question 6

Laws of Logarithms

6. (a) Given that

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2}$$

show that

$$x^3 + 6x^2 + x - 4 = 0 \quad (4)$$

(b) Given also that -1 is a root of the equation

$$x^3 + 6x^2 + x - 4 = 0$$

(i) use algebra to find the other two roots of the equation. (3)

(ii) Hence solve

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2} \quad (1)$$

Worked Solution - Question 6

1. Combine the logarithms

$2\log_4(x+3) = \log_4((x+3)^2)$, so the left side is $\log_4(x(x+3)^2)$.

2. Rewrite one half

$\frac{1}{2} = \log_4 2$, so the right side is $\log_4(4x+2) + \log_4 2 = \log_4(8x+4)$.

3. Equate arguments

Therefore $x(x+3)^2 = 8x+4$.

4. Expand

$x(x^2 + 6x + 9) = 8x + 4$, so $x^3 + 6x^2 + 9x = 8x + 4$.

5. Show the cubic

Rearranging gives $x^3 + 6x^2 + x - 4 = 0$.

6. Use the given root

Since -1 is a root, $(x+1)$ is a factor of $x^3 + 6x^2 + x - 4$.

7. Factor the cubic

$x^3 + 6x^2 + x - 4 = (x+1)(x^2 + 5x - 4)$.

8. Solve the quadratic

$x^2 + 5x - 4 = 0$ gives $x = \frac{-5 \pm \sqrt{25 + 16}}{2} = \frac{-5 \pm \sqrt{41}}{2}$.

9. Apply the logarithm domain

The original log equation requires $x > 0$. The only positive root is

$$x = \frac{-5 + \sqrt{41}}{2}.$$

Final answer

(a) Proven.

$$(b)(i) \ x = \frac{-5 + \sqrt{41}}{2} \text{ and } x = \frac{-5 - \sqrt{41}}{2}.$$

$$(b)(ii) \ x = \frac{-5 + \sqrt{41}}{2}.$$

Question 7

Modelling with Sequences & Series

7. Wheat is grown on a farm.

- In year 1, the farm produced 300 tonnes of wheat.
- In year 12, the farm is predicted to produce 4000 tonnes of wheat.

Model A assumes that the amount of wheat produced on the farm will increase by the same amount each year.

- (a) Using model A , find the amount of wheat produced on the farm in year 4.
Give your answer to the nearest 10 tonnes.

(3)

Model B assumes that the amount of wheat produced on the farm will increase by the same percentage each year.

- (b) Using model B , find the amount of wheat produced on the farm in year 2.
Give your answer to the nearest 10 tonnes.

(3)

- (c) Calculate, according to the two models, the difference between the total amounts of wheat predicted to be produced on the farm from year 1 to year 12 inclusive.
Give your answer to the nearest 10 tonnes.

(3)

Worked Solution - Question 7

Topic group

1. Model A common difference

Model A is arithmetic. From year 1 to year 12 there are **11** equal increases, so

$$d = \frac{4000 - 300}{11} = \frac{3700}{11}.$$

2. Find year 4 in model A

Year 4 is $300 + 3d = 300 + 3\left(\frac{3700}{11}\right) = 1309.09\dots$, so the amount is **1310** tonnes to the nearest **10** tonnes.

3. Model B common ratio

Model B is geometric. If the ratio is r , then $300r^{11} = 4000$.

4. Find the ratio

$$r = \left(\frac{4000}{300}\right)^{1/11} = 1.2655\dots$$

5. Find year 2 in model B

Year 2 is $300r = 379.65\dots$, so the amount is **380** tonnes to the nearest **10** tonnes.

6. Total for model A

$$S_A = \frac{12}{2}(300 + 4000) = 25800 \text{ tonnes.}$$

7. Total for model B

$$S_B = 300 \frac{r^{12} - 1}{r - 1} = 17935.20\dots \text{ tonnes.}$$

8. Find the difference

The difference is $25800 - 17935.20 \dots = 7864.79 \dots$, which is **7860** tonnes to the nearest **10** tonnes.

Final answer

(a) 1310 tonnes.

(b) 380 tonnes.

(c) 7860 tonnes.

Question 8

Proof

8. (i) Use a counter example to show that the following statement is **false**

“ $n^2 + 3n + 1$ is prime for all $n \in \mathbb{N}$ ”

(2)

- (ii) Use algebra to prove by exhaustion that for all $n \in \mathbb{N}$

“ $n^2 - 2$ is **not** a multiple of 4”

(4)

Worked Solution - Question 8

Topic group

1. Choose a counterexample

Take $n = 6$.

2. Evaluate the expression

$$n^2 + 3n + 1 = 6^2 + 3(6) + 1 = 36 + 18 + 1 = 55.$$

3. Conclude false

55 is not prime, so the statement is false.

4. Use residues modulo four

Every natural number is one of $4k$, $4k + 1$, $4k + 2$ or $4k + 3$.

5. Check n equals $4k$

If $n = 4k$, then $n^2 - 2 = 16k^2 - 2 = 4(4k^2 - 1) + 2$, not a multiple of 4.

6. Check n equals $4k$ plus one

If $n = 4k + 1$, then $n^2 - 2 = 16k^2 + 8k - 1 = 4(4k^2 + 2k - 1) + 3$, not a multiple of 4.

7. Check n equals $4k$ plus two

If $n = 4k + 2$, then $n^2 - 2 = 16k^2 + 16k + 2 = 4(4k^2 + 4k) + 2$, not a multiple of 4.

8. Check n equals $4k$ plus three

If $n = 4k + 3$, then $n^2 - 2 = 16k^2 + 24k + 7 = 4(4k^2 + 6k + 1) + 3$, not a multiple of 4.

9. Complete the proof

All possible cases leave remainder **2** or **3** when divided by **4**, so $n^2 - 2$ is not a multiple of **4**.

Final answer

(i) Counterexample: $n = 6$ gives **55**, which is not prime.

(ii) Proven by considering $n = 4k, 4k + 1, 4k + 2, 4k + 3$.

Question 9

Trigonometric Equations

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for
- $0 \leq x < 360^\circ$
- , the equation

$$\sin x \tan x = 5$$

giving your answers to one decimal place.

(6)

- (ii)

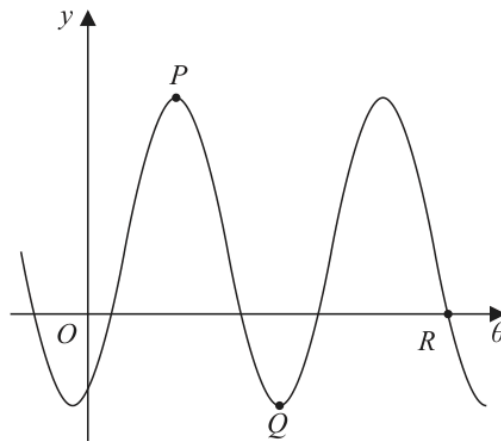


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = A \sin\left(2\theta - \frac{3\pi}{8}\right) + 2$$

where A is a constant and θ is measured in radians.The points P , Q and R lie on the curve and are shown in Figure 1.Given that the y coordinate of P is 7

- (a) state the value of
- A
- ,

(1)

- (b) find the exact coordinates of
- Q
- ,

(3)

- (c) find the value of
- θ
- at
- R
- , giving your answer to 3 significant figures.

(4)

Worked Solution - Question 9

1. Rewrite the equation

$\sin x \tan x = 5$ becomes $\frac{\sin^2 x}{\cos x} = 5$.

2. Use sine squared

Multiplying by $\cos x$ gives $\sin^2 x = 5 \cos x$. Since $\sin^2 x = 1 - \cos^2 x$,
 $1 - \cos^2 x = 5 \cos x$.

3. Solve for cos x

$\cos^2 x + 5 \cos x - 1 = 0$, so $\cos x = \frac{-5 \pm \sqrt{29}}{2}$.

4. Choose the possible cosine value

$\frac{-5 - \sqrt{29}}{2} < -1$, so the valid value is $\cos x = \frac{-5 + \sqrt{29}}{2}$.

5. Find x

For $0 \leq x < 360^\circ$, this gives $x = 78.9^\circ$ or 281.1° to one decimal place.

6. Find A from P

The point P is a maximum point on the sine curve. Its y coordinate is 7 , so
 $A + 2 = 7$ and $A = 5$.

7. Locate Q

Point Q is the next minimum after P , so $\sin \left(2\theta - \frac{3\pi}{8} \right) = -1$.

8. Find theta at Q

For this minimum, $2\theta - \frac{3\pi}{8} = \frac{3\pi}{2}$, so $2\theta = \frac{15\pi}{8}$ and $\theta = \frac{15\pi}{16}$.

9. Find y at Q

At the minimum, $y = 2 - 5 = -3$, so $Q = \left(\frac{15\pi}{16}, -3\right)$.

10. Set up R

At R , $y = 0$, so $5 \sin\left(2\theta - \frac{3\pi}{8}\right) + 2 = 0$.

11. Choose the shown crossing

Thus $\sin\left(2\theta - \frac{3\pi}{8}\right) = -\frac{2}{5}$. The crossing R shown after the second maximum has $2\theta - \frac{3\pi}{8} = 3\pi + \sin^{-1}\left(\frac{2}{5}\right)$.

12. Find theta at R

$\theta = \frac{1}{2}\left(3\pi + \sin^{-1}\left(\frac{2}{5}\right) + \frac{3\pi}{8}\right) = 5.507\dots$, so $\theta = 5.51$ to 3 significant figures.

Final answer

(i) $x = 78.9^\circ, 281.1^\circ$.

(ii)(a) $A = 5$.

(ii)(b) $Q = \left(\frac{15\pi}{16}, -3\right)$.

(ii)(c) $\theta = 5.51$.

Question 10

Applications of Differentiation

10.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

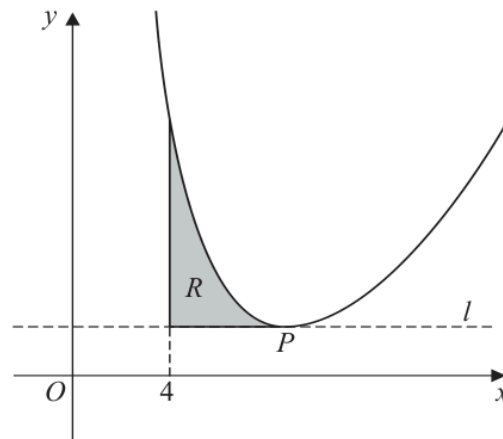


Figure 2

Figure 2 shows a sketch of the curve with equation

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

The point P is the only stationary point on the curve.(a) Use calculus to show that the x coordinate of P is 9

(4)

The line l passes through the point P and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve, the line l and the line with equation $x = 4$ (b) Use algebraic integration to find the exact area of R .

(5)

Worked Solution - Question 10

Topic group

1. Rewrite the curve

$$y = \frac{1}{2}x^2 + 1458x^{-3/2} - 74.$$

2. Differentiate

$$\frac{dy}{dx} = x + 1458 \left(-\frac{3}{2} \right) x^{-5/2} = x - 2187x^{-5/2}.$$

3. Use the stationary condition

At a stationary point, $x - 2187x^{-5/2} = 0$.

4. Solve for x

$$x = 2187x^{-5/2}, \text{ so } x^{7/2} = 2187 = 3^7.$$

5. Show x coordinate

Raising both sides to the power $\frac{2}{7}$ gives $x = 3^2 = 9$. Since $x > 0$, this is the only stationary point.

6. Find the horizontal line

$$\text{At } x = 9, y = \frac{1}{2}(9)^2 + \frac{1458}{\sqrt{9^3}} - 74 = \frac{81}{2} + 54 - 74 = \frac{41}{2}.$$

7. Set up the area

The region lies from $x = 4$ to $x = 9$ between the curve and the horizontal line $y = \frac{41}{2}$.

8. Integrate the difference

$$R = \int_4^9 \left(\frac{1}{2}x^2 + 1458x^{-3/2} - 74 - \frac{41}{2} \right) dx.$$

9. Find an antiderivative

$$\int \left(\frac{1}{2}x^2 + 1458x^{-3/2} - \frac{189}{2} \right) dx = \frac{x^3}{6} - 2916x^{-1/2} - \frac{189}{2}x.$$

10. Evaluate exactly

$$R = \left[\frac{x^3}{6} - 2916x^{-1/2} - \frac{189}{2}x \right]_4^9 = \frac{373}{3}.$$

Final answer

(a) Proven.

(b) $\frac{373}{3}$.